

ISAAC (Zac) ADJEI

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PROFILE

Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026. Builds across the hardware-software stack from PCB-level design and embedded firmware to IoT platforms. Seeking embedded systems and hardware-software integration internships.

EDUCATION

Aston University, Birmingham, United Kingdom

Oct 2024 - Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

Relevant Modules: Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Electronics 1, Electronics 2, Introductory Mathematics for Engineering, Power Skills

- Developing strong foundations in software engineering, electronics and applied mathematics while building practical project experience across embedded systems, AI/data and full-stack development

Stanmore College, London, United Kingdom

Sep 2022 - Jul 2024

Pearson BTEC Level 3 National Extended Diploma in Engineering · **D*DD (Distinction*, Distinction, Distinction)**

Best and Most Hardworking Student Award

Relevant Modules: Engineering Product Design and Manufacture, Microcontroller Systems for Engineers, Engineering Principles, Specialist Engineering Project, CAD in Engineering, Electronic Devices and Circuits, Calculus to Solve Engineering Problems, Electronic Measurement and Testing of Circuits

- Completed intensive hands-on training in engineering design, testing and documentation, applying microcontroller, electronics and CAD skills through practical coursework and projects

SKILLS

Embedded & Hardware: Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, PCB Layout, Circuit Design, Oscilloscope, I2C / UART / SPI, FPGA, VHDL

Languages: C, C++, Python, Java, TypeScript, JavaScript, PHP, SQL, MATLAB

Web and Frameworks: Next.js, React, Node.js, Tailwind CSS, PHP, MySQL, Apache

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Vercel, Cloudflare, GitHub Actions

Professional: Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Time Management, Adaptability, Attention to Detail

AI/ML and Data: TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV, Jupyter Notebooks

PROJECTS

PHAEMOS: Smart Maintenance Platform

2026, ongoing

C++ (ESP32, Arduino), C (STM32 HAL), MicroPython (Pico 2W), FastAPI, Next.js 15,

[Website ↗](#) | [GitHub ↗](#) | [Portfolio ↗](#)

PostgreSQL 15, Redis 7, scikit-learn, Docker, Prometheus, Grafana

- Architected a **4-node IoT platform** (ESP32, STM32 Black Pill, Arduino Nano, Pico 2W/MicroPython) collecting **18 real-time sensor streams** (vibration, IR, gas, RPM, moisture, sound); **FastAPI + PostgreSQL 15 + Redis 7** backend processing telemetry in under **200ms per request**
- **Isolation Forest anomaly detection** on every ingest, auto-generating maintenance tickets with diagnostic recommendations; STM32 runs **100Hz FFT vibration analysis** via ARM CMSIS-DSP (**9x speedup** over naive DFT)
- **JWT refresh flow, Google/GitHub OAuth, TOTP 2FA**, multi-tenant **RBAC**, rate limiting, immutable audit log; **WebSocket** live dashboard; **Prometheus + Grafana** monitoring; deployed on **DigitalOcean VPS + Nginx**

4×4×4 NeoPixel LED Cube

2025

Arduino, C++, WS2812B, Embedded Systems

[GitHub ↗](#) | [Portfolio ↗](#)

- Built a 64-LED interactive display with a **non-blocking state machine architecture** using millis()
- Introduced four animation modes, ambient-light adaptive brightness and debounced user controls

CNC Milling Machine Control System

2024

Arduino, C++, Safety Systems, State Machines

[Portfolio ↗](#)

- Engineered a **safety-focused controller** using **finite state machine logic** and strict transition validation
- Integrated door interlocks, **emergency stop handling**, timed cooldown and LCD status feedback for safe operation

Two-Stage Audio Amplifier

2026

Analogue Design, PCB Design, KiCad, Proteus, Electronics

[Portfolio ↗](#)

- Delivered a **full engineering workflow: simulation, breadboard validation, PCB layout and bench testing**
- Validated performance with measured output of **2.980 Vpp** against a **3 Vpp design target**

AstonCV - Full-Stack CV Database

2026

PHP, MySQL, CSS, Apache, Cloudflare

[Website ↗](#) | [GitHub ↗](#) | [Portfolio ↗](#)

- Engineered **12+ security controls** including **bcrypt hashing**, **CSRF tokens**, **PDO prepared statements** and **brute-force protection**
- **Deployed live** with searchable profiles, sorting/filtering and server-side PDF export

git-unlocked - Open Source Git Course

2026

Git, GitHub, GitLab, Markdown, Open Source

[GitHub ↗](#) | [Portfolio ↗](#)

- Created a structured learning resource with **217+ topic files** covering fundamentals to advanced workflows across Git, GitHub, GitLab and 5 other platforms
- Maintained contributor workflow with issue templates, PR standards and CI checks

EXPERIENCE

Junior Apprentice HVAC Technician

Massive Refrigeration Services, Accra, Ghana

Field servicing, diagnostics, installation support

Jul 2019 - Jul 2021

- Serviced, installed and maintained **50+ air conditioning units**, applying practical knowledge of **electrical systems**, refrigerant circuits, thermodynamics and diagnostic instrumentation in live field environments

Student Representative

Aston Students' Union, Birmingham, UK

Student advocacy, stakeholder communication, academic liaison

Sep 2025 - Present

- Represent Electronic Engineering students, conveying technical module feedback to academic staff through **SSLC, Senate and Council meetings**
- Influenced **electronics and programming curriculum improvements** by gathering cohort feedback and presenting actionable proposals at staff-student forums

Diplomatic, Consular and Estates Intern

Ghana High Commission, London, UK

Consular operations, diplomatic property, document processing, client support

Mar 2024 - Aug 2024

- Operated official consular processing systems handling **biometric capture**, document authentication and physical visa and passport production, supporting **100+ daily client interactions**
- Assisted with **physical site inspections** and facility surveys, maintaining estates and maintenance records and coordinating with officers on operational logistics

RESEARCH & PUBLICATIONS

git-unlocked: An Open-Source Modular Curriculum for Git and Version Control Education

I. Adjei · Zenodo · Jun 2026

[Zenodo ↗](#) | [DOI ↗](#)

- Open-source modular curriculum covering **12 thematic sections** and **217+ topic files** across Git, GitHub and GitLab; published as a citable research artefact with a permanent DOI

VOLUNTEERING

Student Judge - National Emerging Talent Awards 2026

Feb 2026

targetjobs UK (Group GTI)

- Evaluated employer submissions for **Best Placement or Internship Programme** using qualitative and scoring criteria; commended for a thorough, timely and high-quality approach

Fundraising Volunteer - 10 Days of 5K Challenge

Feb - Mar 2026

Cancer Research UK

- Completed **10 × 5km runs (50km+)** for the Cancer Research UK 10 Days of 5K Challenge, managing a fundraising page that contributed to a campaign raising over **£797,424.64** (plus £159,224.14 Gift Aid)

SPOKEN LANGUAGES

English (Full professional), **Twi & Ga** (Full Native), **French & Spanish** (Elementary)